

Revenue Leakage Analysis | October 2024

Exploratory Data Analysis · SQL · Power BI

PROBLEM STATEMENT

The Souled Store processed **60,000 orders** totalling **₹62.46M GMV** in October 2024, yet a meaningful slice of that revenue never converted to profit. This analysis quantifies where value is being lost through cancellations, COD friction, aggressive discounting, and high-return SKUs. Each finding is paired with a specific, number-backed recommendation.

DATASET OVERVIEW

Metric	Value	Metric	Value
Total Orders	60,000	Products	80 SKUs
Gross GMV	₹62.46M	Franchises / Themes	14
Avg Order Value	₹1,041	Period	Oct 1–31, 2024
Cities Covered	Pan-India	Payment Methods	6 (COD, UPI, etc.)

KEY FINDINGS

01 ₹5.35M in Cancelled Order Value Total revenue lost to order cancellations reached ₹5.35M across 5,000 cancelled orders. an 8.54% overall cancellation rate. COD alone accounts for 22.9% of that loss (₹1.23M), despite representing only 15.7% of total order volume.	02 Discount Erosion: 88.6% of Orders Carry a Discount 51,647 of 60,000 orders had a discount applied, with an average discount of 15.3% (₹144.73/order). Total discount outlay hit ₹8.68M. Gross margin held at 31.47%, but 30%+ discount bands show the sharpest margin compression.
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03

Instagram Drives ₹15.9M, 4.2x More Than Direct Traffic

Instagram is the #1 revenue channel at ₹15.9M, followed by Influencer Marketing (₹10.9M) and Facebook Ads (₹9.6M). Organic Search contributes ₹7.1M. Direct Traffic lags at ₹3.8M, signalling low brand-recall shopping behaviour.

04

Heritage Sneaker Low Top at ₹4.85M Tops Revenue, Return Risk Noted

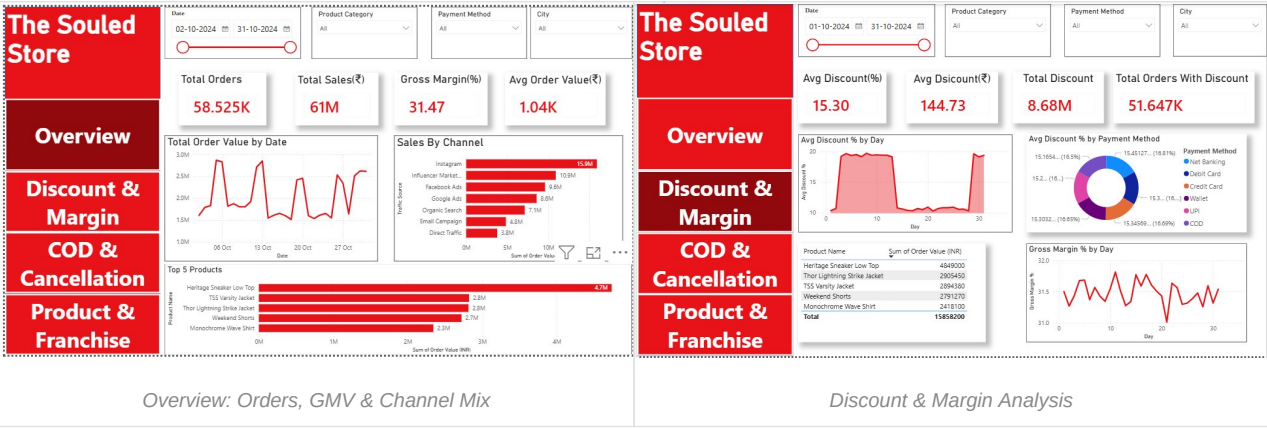
The top revenue SKU (Heritage Sneaker Low Top, ₹4.85M) and Thor Lightning Strike Jacket (₹2.9M) also rank among the highest-volume products. Return rate analysis flags that revenue leaders don't always have the cleanest post-purchase metrics.

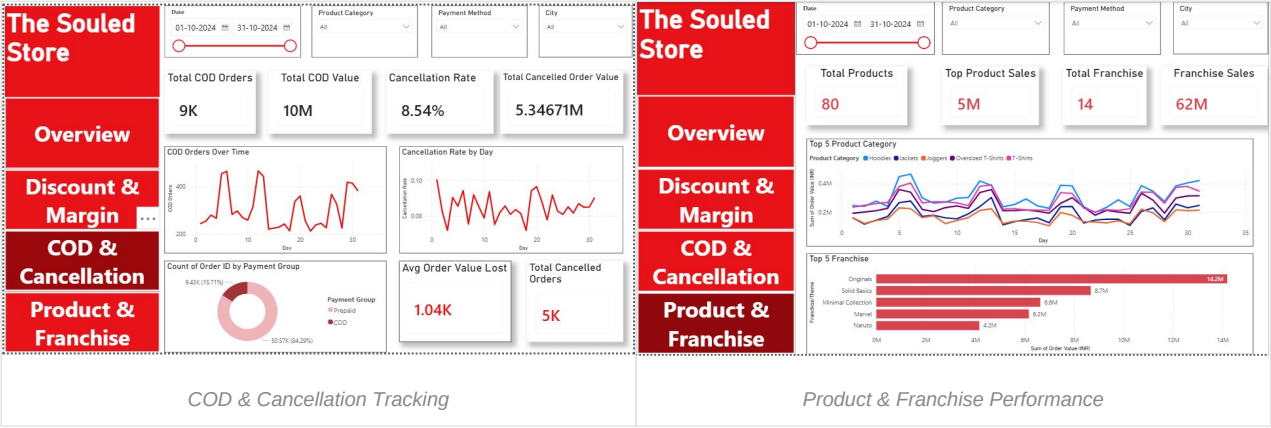
05

Originals Franchise Dominates at ₹14.2M

The Originals franchise leads all themes at ₹14.2M, followed by Solid Basics (₹8.7M) and Minimal Collection (₹6.6M). Marvel and Naruto IP franchises each clear ₹4–6M, confirming licensed content as a consistent revenue driver alongside house brands.

POWER BI DASHBOARD PREVIEW





SQL HIGHLIGHTS

COD Cancellation Rate by City

Identifies which cities have the worst COD fallout. Critical for logistics risk tiering.

```
SELECT City,
       COUNT(*) AS Total_COD_Orders,
       ROUND(100.0 * SUM(CASE WHEN Order_Status = 'Cancelled'
                              THEN 1 ELSE 0 END) / COUNT(*), 2) AS Cancellation_Rate
FROM SOULED_STORE
WHERE Payment_Method = 'COD'
GROUP BY City
ORDER BY Cancellation_Rate DESC;
```

Discount Band vs. Margin

Quantifies margin erosion at each discount tier. The core evidence for the discount cap recommendation.

```
SELECT
  CASE WHEN Discount_Applied = 0 THEN '0%'
        WHEN Discount_Applied BETWEEN 1 AND 10 THEN '1-10%'
        WHEN Discount_Applied BETWEEN 11 AND 20 THEN '11-20%'
        ELSE '20%+' END AS Discount_Band,
  COUNT(*) AS Order_Volume,
  ROUND(AVG(Estimated_Profit_Margin), 2) AS Avg_Margin
FROM SOULED_STORE
GROUP BY Discount_Band ORDER BY Avg_Margin DESC;
```

Delivery Partner Rating & Return Correlation

Links delivery partner performance (speed, rating) to post-delivery return rates. Reveals operational NPS risk.

```
SELECT Delivery_Partner,
       ROUND(AVG(Customer_Rating), 2) AS Avg_Rating,
       ROUND(AVG(Delivery_Time_Days), 2) AS Avg_Delivery_Days,
       ROUND(100.0 * SUM(CASE WHEN Return_Request = 'Yes'
                              THEN 1 ELSE 0 END) / COUNT(*), 2) AS Return_Rate
FROM SOULED_STORE
WHERE Customer_Rating IS NOT NULL
GROUP BY Delivery_Partner
ORDER BY Avg_Rating ASC;
```

RECOMMENDATIONS

01 Restrict COD in High-Cancellation Cities

COD accounts for █1.23M in cancellation losses despite being only 15.7% of orders. Introduce a city-level COD eligibility score: restrict or add a nominal prepayment deposit (█50–100) for high-COD-cancellation pin codes. This alone could recover █400–600K annually with minimal checkout friction.

02 Set a 20% Discount Cap on High-Value SKUs

Originals franchise drives █14.2M in revenue and is the brand's most elastic property. The current 15.3% average discount with █8.68M total outlay suggests discounting is being applied bluntly rather than strategically. Reserve 21%+ discounts for clearance inventory and end-of-season events only. This protects the current 31.47% gross margin baseline.

03 Scale Instagram and Influencer Spend; Review Google Ads ROI

Instagram (█15.9M) and Influencer Marketing (█10.9M) together drive 43% of GMV. Facebook Ads (█9.6M) and Google Ads (█8.6M) add volume but at higher paid spend. Direct Traffic (█3.8M) indicates low brand loyalty. Invest in post-purchase CRM flows (email/WhatsApp) to convert first-time buyers into repeat customers, especially given the █1,041 AOV opportunity.

TOOLS USED

Tool	Purpose	Key Output
Python (pandas, seaborn)	Exploratory Data Analysis & visualisation	EDA notebook with 15+ charts
SQL (Standard)	Aggregation, window functions, cohort queries	5 analytical queries for business KPIs
Power BI	Interactive dashboard across 4 analytical pages	Filterable report with slicers + KPI cards

Dataset is synthetic and self-generated for portfolio purposes. All figures, order records, and customer data are fictional and do not represent actual Souled Store operations or financials. Prepared as part of a Data Analyst application.